

Common-Mode Voltage Reduction in Two-Level Inverters by Introducing Adaptive Carrier Phase Shifts

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Abstract—Voltage source inverters (VSIs) are commonly used in industrial motor drive applications, but common-mode voltage (CMV) poses challenges such as electromagnetic interference (EMI) and bearing currents, compromising the reliability of the system. On the one hand, multi-level topologies or common mode filters can mitigate these challenges, but they introduce added complexity and cost. On the other hand, modulation-based methods offer a simpler solution without hardware changes. This study proposes an adaptive carrier-phase-shift technique for sinusoidal pulse width modulation (SPWM) that dynamically adjusts the phase shift within each switching interval to reduce CMV more effectively. To validate the method, an experimental setup was established, and the results were compared with analytical calculations. The proposed approach significantly reduces CMV, improving system reliability and EMI performance while maintaining the simplicity of two-level VSIs.

Index Terms—modulation, common-mode voltage, 2-Level inverter, carrier-phase-shift

I. INTRODUCTION

Voltage source inverters (VSIs) play a significant role in industrial motor-drive applications [1]. Among these, two-level VSIs remain the preferred choice over more complex topologies due to their simple structure, reduced component count, and high reliability [2]. Also, the emergence of wide-bandgap (WBG) devices like silicon carbide (SiC) and gallium nitride (GaN) has further improved the performance of two-level VSIs, enabling higher switching frequencies, reduced losses, and enhanced overall efficiency [3]. Consequently, the mature manufacturing processes for two-level VSIs and the help of the new WBG devices on the performance provide a robust foundation, making the shift to newer topologies less appealing for many industrial applications.

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However, common-mode voltage (CMV) issues arising from switching and modulation are inevitable in VSIs. The presence of CMV can lead to electromagnetic interference (EMI) and cause common-mode currents to flow through the motor chassis or bearings [4], [5]. These currents can accelerate the degradation of bearings and insulation, leading to fatigue faults and reducing system reliability [6]. As a result, mitigating CMV is critical for enhancing the reliability of motor-drive systems.

Various methods to mitigate CMV have been explored, such as new topologies, filter designs, and innovative modulation strategies [7]. While multi-level or modified two-level topologies can reduce CMV, they come with increased cost and complexity [8]–[12]. In contrast, modulation-based solutions offer a practical alternative by reducing CMV with minimal hardware modifications, requiring little additional space, and maintaining system simplicity. This study focuses on modulation-based techniques for CMV mitigation [13]–[19].

Several studies have explored carrier-phase shift techniques to mitigate common-mode voltage (CMV). The authors in [13] introduced interleaved carriers (0 , $-\frac{2\pi}{3}$, and $\frac{2\pi}{3}$) in three-phase SPWM, aiming to redistribute harmonics into positive, negative, and zero sequences. This approach was later extended to multiphase drives, including five-phase and both symmetrical and asymmetrical six-phase systems [14], as well as to three-level inverters [15]. Alternating carrier phase shift methods, which apply 0° or 180° shifts, were introduced for third harmonic injection SPWM, dividing the fundamental period into six segments and adjusting carrier phases accordingly [16]. A sector-based carrier phase shift approach was later employed in high-frequency SiC-based VSIs [17]. Further advancements include a sampling-time adaptive multi-carrier PWM algorithm, where four different carrier phase shift sets are evaluated at each sampling instant to minimize CMV [18]. More recently, a generalized multicarrier

PWM technique was introduced, employing variable carrier phase shifts for phase-B and phase-C, optimizing both CMV and total harmonic distortion (THD) [19].

In all the aforementioned studies, the carrier phase shift either remains constant throughout a fundamental period or its polarity is adjusted based on the voltage sector to achieve the desired reference voltage. While these methods effectively reduce the average CMV over a fundamental period, further improvement can be achieved by minimizing CMV at each switching interval. To address this, we propose an adaptive carrier phase shift approach for SPWM, where the phase shift dynamically adjusts within each switching interval. This method aims to further reduce instantaneous CMV, enhancing overall CMV mitigation beyond the capabilities of conventional fixed or sector-based phase shift techniques.

The paper is organized as follows. Section II analyzes the SPWM and common mode voltages in the 2-Level inverter. Section III explores the conventional interleaved multi-carrier SPWM. Section IV proposes and analyzes the adaptive carrier phase shift method to reduce the CMV. Section VI simulates the proposed modulation method and conventional methods to compare the performance regarding CMV reduction. Section V gives the experimental validation of the proposed method. Section VI concludes and provide possible future works.

II. ANALYTICAL ANALYSIS OF COMMON MODE VOLTAGE FOR SPWM

The common mode voltage in 2-Level 3-phase VSI, shown in Fig. 1, is calculated by

$$V_{CM} = \frac{1}{3}(V_A + V_B + V_C). \quad (1)$$

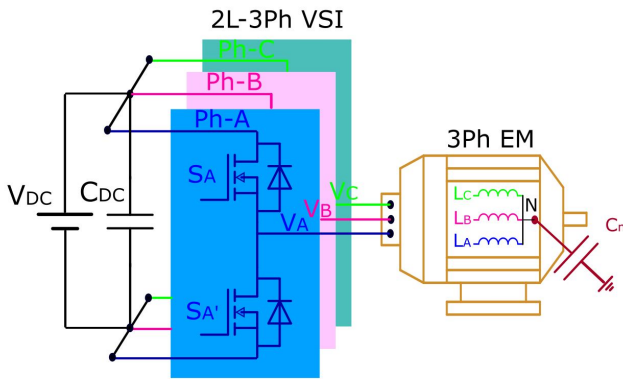


Fig. 1. The system structure of a 2-level 3-phase motor drive system.

The CMV leads to a common-mode current through motor shafts and bearings which can be modeled by the capacitor (C_n) connected between the neutral point and ground as given in Fig. 1. When conventional single-carrier SPWM is implemented, the phase-to-neutral voltages (V_{p-n}) in the

frequency domain can be calculated by using a Double Fourier series as:

$$V_{p-n} = \frac{V_{DC}}{2} + m_a \frac{V_{DC}}{2} \cos(\omega_R t + \theta_R) + \frac{2V_{DC}}{\pi} \sum_{m=1}^{\infty} J_0\left(m \frac{\pi}{2} m_a\right) \sin\left(m \frac{\pi}{2}\right) \cos\left(m(\omega_C t + \theta_C)\right) + \frac{2V_{DC}}{\pi} \sum_{m=1}^{\infty} \sum_{n=-\infty}^{\infty} \left[\frac{1}{m} J_n\left(m \frac{\pi}{2} m_a\right) \sin\left((m+n) \frac{\pi}{2}\right) \cos\left(m(\omega_C t + \theta_C) + n(\omega_R t + \theta_R)\right) \right] \quad (2)$$

where ω_R , ω_C , θ_R , θ_C , are angular frequencies and phases of reference and carrier signals, and J_0 , J_k are the zeroth and k^{th} order Bessel functions. The key components of the switching function, or leg voltage, include: the DC component, the baseband harmonics, the carrier harmonics, and the sideband harmonics. These harmonic components create positive, negative, and zero sequences regarding the phase shift of the carrier and reference signals. While positive and negative sequence voltage harmonics create differential-mode currents, zero-sequence voltage harmonics create common-mode currents. For the conventional SPWM with a single carrier signal, the common mode voltage harmonics can be calculated by using (1) and (2) as:

$$V_{CM} = \frac{V_{DC}}{2} + \frac{2V_{DC}}{\pi} \sum_{\substack{m=1 \\ m \text{ is odd}}}^{\infty} \left[\frac{J_0\left(m \frac{\pi}{2} m_a\right) \sin\left(m \frac{\pi}{2}\right)}{\cos(m\omega_C t)} \right] + \frac{2V_{DC}}{\pi} \sum_{\substack{m=1 \\ m \equiv 0 \pmod{3}}}^{\infty} \sum_{\substack{n=-\infty \\ (m+n) \text{ is not even}}}^{\infty} \left[\frac{1}{m} J_n\left(m \frac{\pi}{2} m_a\right) \sin\left((m+n) \frac{\pi}{2}\right) \cos\left(m(\omega_C t) + n(\omega_R t)\right) \right] \quad (3)$$

Therefore, the common-mode harmonics appear at odd carrier harmonics, at odd triplen harmonics near the even carrier harmonics, and at even triplen harmonics near the odd carrier harmonics. The sequences for the first carrier with sideband harmonics are presented in Table I.

TABLE I
FREQUENCY COMPONENTS AND THEIR SEQUENCES AROUND FIRST CARRIER HARMONIC.

Frequency	Phase-A	Phase-B	Phase-C	Sequence
f_0	0°	120°	-120°	Positive
$f_s - 6f_0$	0°	0°	0°	Zero
$f_s - 4f_0$	0°	-120°	120°	Negative
$f_s - 2f_0$	0°	120°	-120°	Positive
f_s	0°	0°	0°	Zero
$f_s + 2f_0$	0°	-120°	120°	Negative
$f_s + 4f_0$	0°	120°	0°	Positive
$f_s + 6f_0$	0°	0°	0°	Zero

Further, an example of an SPWM operation point with a single carrier signal is given in Fig. 2, to show the common mode voltage.

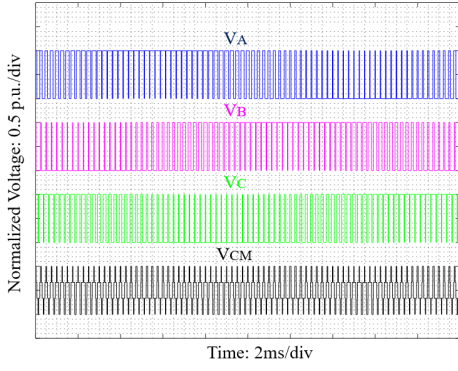


Fig. 2. The normalized common mode voltage for $m_a = 0.85$.

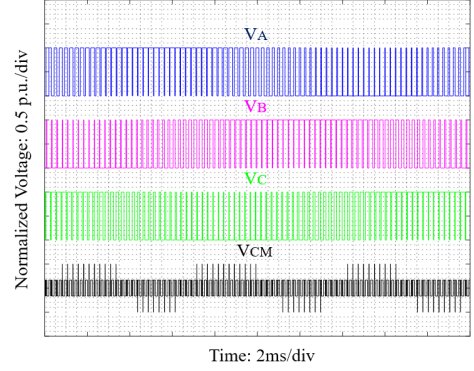


Fig. 3. The normalized common mode voltage for $m_a = 0.85$.

III. THE CONVENTIONAL INTERLEAVED MULTI-CARRIER SPWM

Using multi-carriers instead of a single carrier, the sequences of harmonics can be alternated by adjusting the phases of each carrier signal. The phases of the frequency component can be calculated using (2). The phases of the first carrier signal with sideband harmonics are given in Table II.

TABLE II
FREQUENCY COMPONENTS AND THEIR PHASES AROUND FIRST CARRIER HARMONIC FOR MULTI-CARRIER SIGNALS.

Frequency	Phase-A	Phase-B	Phase-C	Sequence
f_0	0°	120°	$-120^\circ + \phi_{CC}$	Positive
$f_s - 6f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative
$f_s - 4f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$-120^\circ + \phi_{CB} = 120^\circ$	$120^\circ + \phi_{CC} = -120^\circ$	Positive
$f_s - 2f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$120^\circ + \phi_{CB} = 0^\circ$	$-120^\circ + \phi_{CC} = 0^\circ$	Zero
f_s	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative
$f_s + 2f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$-120^\circ + \phi_{CB} = 120^\circ$	$120^\circ + \phi_{CC} = -120^\circ$	Positive
$f_s + 4f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$120^\circ + \phi_{CB} = 0^\circ$	$0^\circ + \phi_{CC} = 0^\circ$	Zero
$f_s + 6f_0$	$0^\circ + \phi_{CA} = 0^\circ$	$0^\circ + \phi_{CB} = -120^\circ$	$0^\circ + \phi_{CC} = 120^\circ$	Negative

$\phi_{CA} = 0$, $\phi_{CB} = -120^\circ$, and $\phi_{CC} = +120^\circ$.

Interleaving the carrier waves of the three phases is applied by shifting them by one-third of the switching period. Here the main idea is to change the zero sequences with positive-negative sequences. The magnitude of the first carrier harmonic is greater than sidebands, therefore altering the zero sequences to sidebands reduces the common mode voltage. Further, an example of an SPWM operation point with multi-carrier signals is given in Fig. 3, to show the change in common mode voltage.

IV. THE PROPOSED ADAPTIVE CARRIER PHASE SHIFT METHOD

The proposed method introduces a variable carrier phase shift to achieve lower common-mode voltage (CMV) during each switching interval. Unlike conventional multi-carrier methods, where the phase shift is kept constant (or only the polarity is altered), this method adaptively adjusts the carrier phase shift to minimize CMV. In multi-carrier sinusoidal pulse width modulation (SPWM), the primary idea is to alter the sequence of the first carrier harmonic, which has a greater magnitude than the sidebands. The proposed method also aims

to reduce the first switching harmonic in the common mode but achieves this by applying an adaptive variable carrier phase shift in each switching period.

For each switching interval, the magnitude of the first switching component can be calculated using the Fourier series as follows:

$$\begin{aligned} V_A(t)^{\omega_c} &= \frac{2V_{DC}}{\pi} \sin(\pi D_A) \cos(\omega_c t + \phi_{CA}), \\ V_B(t)^{\omega_c} &= \frac{2V_{DC}}{\pi} \sin(\pi D_B) \cos(\omega_c t + \phi_{CB}), \\ V_C(t)^{\omega_c} &= \frac{2V_{DC}}{\pi} \sin(\pi D_C) \cos(\omega_c t + \phi_{CC}), \end{aligned} \quad (4)$$

where the duty cycles D_A , D_B , and D_C follow the reference sinusoidal waveforms:

$$\begin{aligned} D_A &= \frac{1 + m_a \cos(\omega_R t)}{2}, \\ D_B &= \frac{1 + m_a \cos(\omega_R t + 120^\circ)}{2}, \\ D_C &= \frac{1 + m_a \cos(\omega_R t - 120^\circ)}{2}. \end{aligned} \quad (5)$$

The common-mode voltage for the first carrier harmonic is then calculated within a switching interval as:

$$V_{CM}^{\omega_c}(t) = \frac{1}{3} [V_A(t)^{\omega_c} + V_B(t)^{\omega_c} + V_C(t)^{\omega_c}]. \quad (6)$$

Thus, we have three signals ($V_A(t)^{\omega_c}$, $V_B(t)^{\omega_c}$, and $V_C(t)^{\omega_c}$) with the same frequency but different magnitudes and phases. This allows us to analyze the CMV in the phasor domain, as illustrated in Fig. 6.

The goal is to minimize $V_{CM}^{\omega_c}$. The three voltage vectors have magnitudes that vary with time and modulation index, as shown in Fig. 5.

If these three vectors form a triangle, their sum can be minimized to zero. This condition can be verified using the triangle inequality. If the vectors do not form a triangle, the minimum CMV can be achieved by aligning the smaller vectors in opposition to the larger vector. In the first case (vectors forming a triangle), the carrier phase shifts can be

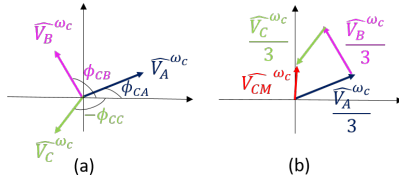


Fig. 4. Phasor domain analysis of the CMV in the first switching harmonic. a) The phase voltages. b) The CMV.

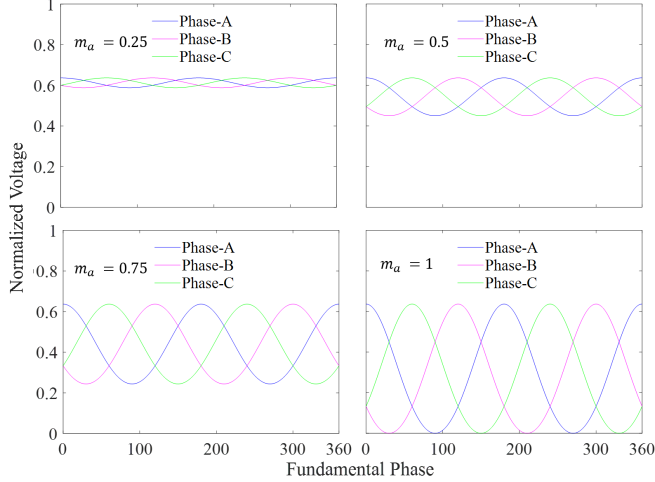


Fig. 5. Magnitude of vectors over a fundamental period for different modulation indexes.

determined using the cosine and sine theorems. In the second case (vectors not forming a triangle), the phases of the smaller vectors are set to 180° relative to the larger vector, effectively canceling its contribution to the CMV.

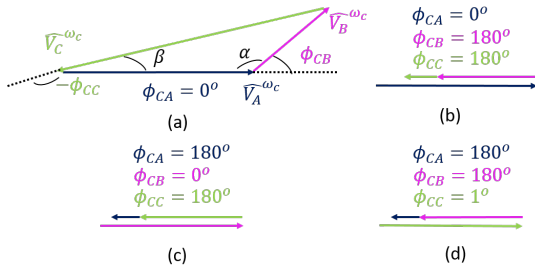


Fig. 6. Determination of the carrier phase shifts in phasor domain analysis. a) Triangle formation. b) Phase-A voltage vector is bigger than other vectors. c) Phase-B voltage vector is bigger than other vectors. d) Phase-C voltage vector is bigger than other vectors.

To make the calculation easy, ϕ_{CA} is taken as zero in triangle formation, and the other two carrier phase shifts can be calculated using the Algorithm 1.

Further, an example of an SPWM operation point with adaptive carrier phase shift method with applied phase shifts are shown in Fig. 7, to show the change in common mode voltage.

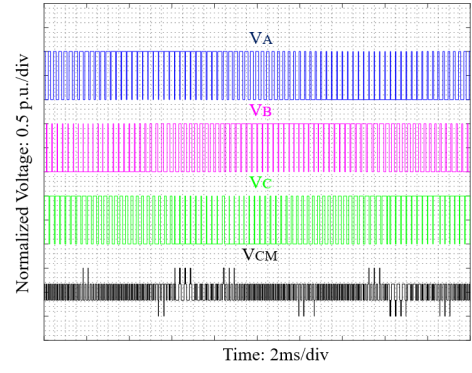


Fig. 7. The normalized common mode voltage for $m_a = 0.85$.

Algorithm 1 Determination of the Carrier Phase Shifts

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1: if  $|\hat{V}_A^{\omega_c} - \hat{V}_B^{\omega_c}| < \hat{V}_C^{\omega_c} < |\hat{V}_A^{\omega_c} + \hat{V}_B^{\omega_c}|$  then
2:   Take  $\varphi_{CA} = 0^\circ$ :
3:   Apply cosine theorem to find  $\varphi_{CB}$ :

$$(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - 2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}\cos(\alpha) = (\hat{V}_C^{\omega_c})^2$$


$$\cos[\alpha] = \frac{(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - (\hat{V}_C^{\omega_c})^2}{2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}}$$


$$\alpha = \arccos\left[\frac{(\hat{V}_A^{\omega_c})^2 + (\hat{V}_B^{\omega_c})^2 - (\hat{V}_C^{\omega_c})^2}{2\hat{V}_A^{\omega_c}\hat{V}_B^{\omega_c}}\right]$$


$$\varphi_{CB} = 180^\circ - \alpha$$

4:   Apply sine theorem to find  $\varphi_{CC}$ :

$$\frac{\sin[\beta]}{\hat{V}_B^{\omega_c}} = \frac{\sin[\alpha]}{\hat{V}_C^{\omega_c}}$$


$$\sin[\beta] = \frac{\hat{V}_B^{\omega_c} \sin[\alpha]}{\hat{V}_C^{\omega_c}}$$


$$\beta = \arcsin\left[\frac{\hat{V}_B^{\omega_c} \sin[\alpha]}{\hat{V}_C^{\omega_c}}\right]$$


$$\varphi_{CC} = \beta - 180^\circ$$

5: else
6:   if  $[\hat{V}_A^{\omega_c} > \hat{V}_B^{\omega_c}]$  and  $[\hat{V}_A^{\omega_c} > \hat{V}_C^{\omega_c}]$  then
7:      $\varphi_{CA} = 0^\circ$ ,  $\varphi_{CB} = 180^\circ$  and  $\varphi_{CC} = 180^\circ$ 
8:   else if  $[\hat{V}_B^{\omega_c} > \hat{V}_A^{\omega_c}]$  and  $[\hat{V}_B^{\omega_c} > \hat{V}_C^{\omega_c}]$  then
9:      $\varphi_{CA} = 180^\circ$ ,  $\varphi_{CB} = 0^\circ$  and  $\varphi_{CC} = 180^\circ$ 
10:  else if  $[\hat{V}_C^{\omega_c} > \hat{V}_A^{\omega_c}]$  and  $[\hat{V}_C^{\omega_c} > \hat{V}_B^{\omega_c}]$  then
11:     $\varphi_{CA} = 180^\circ$ ,  $\varphi_{CB} = 180^\circ$  and  $\varphi_{CC} = 0^\circ$ 
12:  end if
13: end if

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V. SIMULATION RESULTS

The simulation study compares three modulation methods:

- Single-carrier SPWM.
- Interleaved multi-carrier SPWM with phase shifts of $\phi_{CA} = 0^\circ$, $\phi_{CB} = -120^\circ$, and $\phi_{CC} = 120^\circ$.
- Proposed adaptive multi-carrier SPWM.

Detailed simulation results will be presented in the full paper.

VI. EXPERIMENTAL VALIDATION

An experimental setup, as illustrated in Fig. 8, has been established to validate the proposed methodology. The setup uses Imperix SiC power module PEB8024 half-bridge configurations to achieve a 2-Level 3-Phase inverter. A 4-channel Teledyne T3DSO1104 oscilloscope is employed for signal acquisition. For voltage measurements, Tektronix THDP0100 differential probes are utilized. The power supply unit is an EA-PS 9500-10 T. The Imperix B-Box RCP controller is implemented to generate and execute the proposed modulation strategy, enabling real-time control.

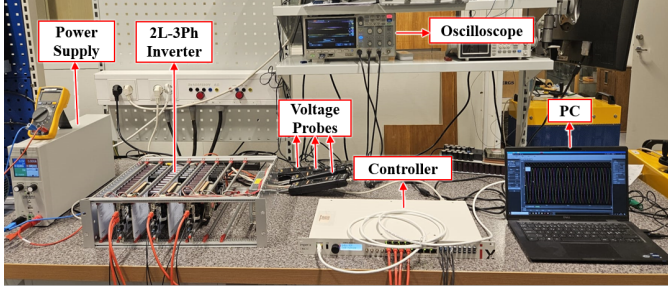


Fig. 8. Experimental setup of 2-Level 3-Phase Inverter System.

To validate the proposed method, the phase voltages for two different modulation indexes ($m_a = 0.25$, $m_a = 0.75$) have been measured, shown in Fig. ??.

VII. FUTURE WORKS AND CONCLUSION

This study proposed an adaptive carrier phase shift technique for sinusoidal pulse width modulation (SPWM) in two-level voltage source inverters (VSIs) to effectively reduce common-mode voltage (CMV). Unlike conventional methods that use fixed or sector-based phase shifts, the proposed approach dynamically adjusts the carrier phase shift within each switching interval, leading to a significant reduction in instantaneous CMV. The analytical analysis and experimental validation demonstrated that the proposed method enhances CMV mitigation, thereby improving system reliability and electromagnetic interference (EMI) performance. The simplicity of the two-level VSI structure is maintained, making this approach a practical solution for industrial motor-drive applications. Future work could explore the application of this technique in multi-phase and multi-level inverters and further optimization for different operating conditions.

REFERENCES

- [1] B. K. Bose, "Power electronics and motor drives recent progress and perspective," *IEEE Transactions on Industrial Electronics*, vol. 56, no. 2, pp. 581–588, 2009.
- [2] I. Husain, B. Ozpıneci, M. S. Islam, E. Gurpinar, G.-J. Su, W. Yu, S. Chowdhury, L. Xue, D. Rahman, and R. Sahu, "Electric drive technology trends, challenges, and opportunities for future electric vehicles," *Proceedings of the IEEE*, vol. 109, no. 6, pp. 1039–1059, 2021.
- [3] J. Millán, P. Godignon, X. Perpiñà, A. Pérez-Tomás, and J. Rebollo, "A survey of wide bandgap power semiconductor devices," *IEEE Transactions on Power Electronics*, vol. 29, no. 5, pp. 2155–2163, 2014.
- [4] F. Fan, K. Y. See, J. K. Banda, X. Liu, and A. K. Gupta, "Investigation and mitigation of premature bearing degradation in motor drive system," *IEEE Electromagnetic Compatibility Magazine*, vol. 8, no. 1, pp. 75–81, 2019.
- [5] J.-K. Park, S.-H. Rhyu, and J. Hur, "Shaft-to-frame voltage mitigation method by changing winding-to-rotor parasitic capacitance of ipmsm," in *2017 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2017, pp. 3571–3576.
- [6] S. Lu, Q. He, T. Yuan, and F. Kong, "Online fault diagnosis of motor bearing via stochastic-resonance-based adaptive filter in an embedded system," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 47, no. 7, pp. 1111–1122, 2017.
- [7] W. Zhu, D. De Gaetano, X. Chen, G. W. Jewell, and Y. Hu, "A review of modeling and mitigation techniques for bearing currents in electrical machines with variable-frequency drives," *IEEE Access*, vol. 10, pp. 125 279–125 297, 2022.
- [8] S. Payami, R. K. Behera, A. Iqbal, and R. Al-Ammari, "Common-mode voltage and vibration mitigation of a five-phase three-level npc inverter-fed induction motor drive system," *IEEE Journal of Emerging and Selected Topics in Power Electronics*, vol. 3, no. 2, pp. 349–361, 2015.
- [9] K. J. Dagan, A. Zuckerberger, and R. Rabinovici, "Fourth-arm common-mode voltage mitigation," *IEEE Transactions on Power Electronics*, vol. 31, no. 2, pp. 1401–1407, 2016.
- [10] A. Salem, M. A. Abido, and F. Blaabjerg, "Common-mode voltage mitigation of dual t-type inverter drives using fast mpc approach," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 8, pp. 7663–7674, 2022.
- [11] S. Xu, Y. Ma, P. Wang, Z. Liu, D. Jiang, Z. Lin, and C. Chen, "Paralleled inverters to drive double dual-three-phase induction motors with common-mode voltage mitigation for traction application," *IEEE Transactions on Transportation Electrification*, vol. 9, no. 2, pp. 3475–3486, 2023.
- [12] F. Acosta-Cambranis, J. Zaragoza, N. Berbel, G. J. Capella, and L. Romeral, "Common-mode voltage mitigation strategies using sigma-delta modulation in five-phase vsis," *IEEE Transactions on Power Electronics*, vol. 37, no. 10, pp. 11 662–11 672, 2022.
- [13] J. W. Kimball and M. Zawodniok, "Reducing common-mode voltage in three-phase sine-triangle pwm with interleaved carriers," *IEEE Transactions on Power Electronics*, vol. 26, no. 8, pp. 2229–2236, 2011.
- [14] N. K. G and S. Srinivas, "Carrier phase shifted spwm for cmv reduction in a three-level inverter using open-end winding induction motor drive," in *2016 IEEE Region 10 Conference (TENCON)*, 2016, pp. 707–712.
- [15] Z. Liu, Z. Zheng, S. D. Sudhoff, C. Gu, and Y. Li, "Reduction of common-mode voltage in multiphase two-level inverters using spwm with phase-shifted carriers," *IEEE Transactions on Power Electronics*, vol. 31, no. 9, pp. 6631–6645, 2016.
- [16] B. Tan, Z. Gu, K. Shen, and X. Ding, "Third harmonic injection spwm method based on alternating carrier polarity to suppress the common mode voltage," *IEEE Access*, vol. 7, pp. 9805–9816, 2019.
- [17] J. Xu, J. Han, Y. Wang, M. Ali, and H. Tang, "High-frequency sic three-phase vsis with common-mode voltage reduction and improved performance using novel tri-state pwm method," *IEEE Transactions on Power Electronics*, vol. 34, no. 2, pp. 1809–1822, 2019.
- [18] A. M. Alcaide, H. Yan, X. Wang, J. I. Leon, R. Portillo, G. Buticchi, S. Vazquez, V. G. Monopoli, M. Liserre, and L. G. Franquelo, "Common-mode voltage mitigation technique in motor drive applications by applying a sampling-time adaptive multi-carrier pwm method," *IEEE Access*, vol. 9, pp. 56 115–56 126, 2021.
- [19] A. M. Alcaide, V. G. Monopoli, E. Zafra, G. Buticchi, J. I. Leon, S. Vazquez, M. Liserre, and L. G. Franquelo, "Generalized multicarrier pwm technique for two-level voltage source inverters," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 5, pp. 4345–4355, 2023.